U209: jointly compatible decoder reductions for the consistency of Peano arithmetic

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Abstract

We compare the three supplied 211-state candidates for the consistency of Peano arithmetic and construct an explicit 209-working-state binary, single-tape candidate, including its blank initializer. The supplied tables are distinct: they delete native states 180, 181, and 218, respectively, from one common 212-state parent. Their patches conflict, so the savings cannot simply be added. A new common source layout makes all three deletions compatible. Starting from the Contextual 211-state candidate, 24 closed source-tree reassociations and two further local folds produce the delivered table, with 418 transitions. The source retains all 3,909 primitive occurrences and all 5,083 ordinal control edges. An independent local proof uses arbitrary unread tape exteriors; a separate full symbolic replay checks 5,084 primitive contracts over arbitrary natural registers, header gaps, and nonempty finite spare lists. The exact 48,140-step blank startup is unchanged. Composing with the supplied numerical-calculus and unary-theory calibration gives the intended PA-formalizable equivalence $\neg\,\mathrm{Halt}(U209, \mathrm{blank}) \leftrightarrow \mathrm{Con}(\mathsf{PA})$. The new release contains mathematical proofs and executable evidence, not a newly exported proof-assistant certificate for this endpoint.

1 Result, conventions, and exact scope

The primary artifact is `machine/u209.tm`. Its working states are $0, \ldots, 208$; its separate halting label is 209. The alphabet is $\{0, 1\}$, the tape is indexed by $\mathbb{Z}$, and the machine starts in state 0 with its head at coordinate 0 on the all-zero tape. Every working-state/symbol pair has exactly one transition, with a left or right move. Initialization is included in the 209-state count.

A row `q b w D r` means: in state q reading b, write w, move one cell in direction D, and enter r. A transition to the halting label performs its write and move before halting. The header `#! start 0` is not a transition. Sparse native labels, including the working state named 211 and halt named 254, are used in the proofs. The exact native/dense bijection is in `machine/state_map.json`.

Property	Delivered value
Working states / binary transition rows	209/418
Source primitive occurrences / distinct subtrees	3,909/197
Resolved ordinal control edges	5,083
Maximum source address depth / zero run	42/6
New rotations relative to Contextual U211	24
States removed relative to common U212	180, 181, 218
Changed retained rows relative to U212 / Contextual	38/26
Source addresses changed relative to U212	3,096

The dense table has SHA-256

`65ccb606c8e970adec22c311486e747b440171363601b8f11b56bc729cf8decb.`

The native table has SHA-256

`891455876f2684b2bd2a9118092715590028fbdb227aa7055030536714d8b7a4.`

These hashes, the full transition table in Appendix B, and the deterministic builders identify the candidate independently of prose.

There are three levels of claim. First, finite exact identities prove that the local folds preserve halting on every common initial tape. Second, source control-flow preservation, quantified primitive contracts, and exact startup connect the address-changing construction to the inherited numerical proof search. Third, the existing interpretation and finite-refutation arguments connect that search to Con(PA). The third level is a retained mathematical dependency, not an assumption that PA is consistent. None of these statements asserts unconditional nonhalting, global minimality, or independently ratified priority.

2 Comparison of the three supplied 211-state proposals

We retrieved the complete incoming directory at ProveIt commit `da4ab9dd68ccfc47ac852118601eaf1fdae53795`. It contains the unary theory archive and the three U211 archives listed below. Their own historical comparison used an earlier eight-packet U212 snapshot at `9d50bf3df8c9e14941b549945c4f8400172d81f4`. These are different comparison sets. The current task concerns three distinct U211 tables. All four input ZIPs, their exact retrieval URLs, sizes, and SHA-256 hashes are retained in `inputs/` [1, 2, 3, 4].

Property	Contextual	Research	Rotated
Closed rotations from U212	6	9	11
Deleted native state	180	181	218
Representative	163	220	157
Changed retained rows	16	19	19
Changed source addresses	549	936	3,096
Maximum address depth	42	42	41
Maximum zero run	6	6	6
Fresh symbolic inference applications	1,277,525	1,280,108	1,245,944

Every proposal has 211 working states, 422 rows, 3,909 source occurrences, 197 distinct source subtrees, and the same ordinal source control-flow graph. They are pairwise non-isomorphic as rooted deterministic transition graphs when start, halt, read symbols, write symbols, and directions are preserved. Since every working state is graph-reachable, the attempted rooted isomorphism is forced; the independent checker records a contradiction for each pair. Graph reachability does not claim that every state is visited in the actual blank computation.

Contextual. Six rotations prepare two states with a common read-one branch and opposite predecessor bits. State 180 is merged into 163; on read zero, the merged state moves left to helper 211. This helper reads the already written predecessor bit and reconstructs the required branch. A separate arithmetic entry at 171/0 must be repaired from 0 L 7 to 0 L 205 because state 7 also serves an interpreter role. The package gives an explicit local proof and a negative control for this repair. Its independent C++ finite executor is particularly useful for new constructions.

Research / Decoder Reassociation. Nine rotations reshape a closed 936-primitive region. State 181 is merged into 220, using helper 129 and different predecessor bits. The four local cases and twelve mutation controls are clearly separated from source semantics. Its source auditor independently re-executes all 3,908 internal prefix bindings at each stage. This is a useful additional guard against incomplete source-to-state bookkeeping.

Rotated. Eleven rotations move a 78-primitive continuation through a larger closed region until every incoming transition to 218 writes zero and moves right. Existing state 157 then reproduces 218 in three steps. The resulting local proof is the simplest of the three: two cases, one invariant, and no change to the representative's own transitions. Its final source depth is 41. Its lower symbolic-inference count is a property of the verification trace, not a demonstrated speed advantage for the unbounded blank computation.

Fresh verification and shared dependencies. All three passed fresh reconstruction, local, source, startup, and full symbolic checks. Each replay regenerated all 5,084 cases, covered every one of its 422 rows, and reproduced the frozen case data exactly. The same independent C++ executor passed 15,252 finite primitive cases for each table. Thus the comparison does not rely only on old PASS flags. Nevertheless, the universal engines share the same inherited inference implementation, and the source/calculus/theory calibration is common. These are three constructions, not three independently developed formal foundations.

Why a joint construction is research rather than patch union. The Contextual/Research, Contextual/Rotated, and Research/Rotated literal patches conflict on five, four, and seven retained rows, respectively. In particular, combining the first two layouts puts both candidate states behind a predecessor one, destroying the bit that the selector needs. The separate deleted labels cannot be added as independent savings. A new layout must satisfy all relevant incoming-bit requirements at once and survive the interpreter's source-address constraints.

3 The new common layout

The primary construction starts with the exact Contextual U211 table and its source. It keeps that candidate's deletion of 180, its helper 211, and its arithmetic entry repair. The new recipe has three phases:

1. Apply the first eight Research rotations. The ninth, involving $(216, 220)$, is already present in the Contextual layout.
2. Apply four further rotations to restore a zero predecessor for 220 while retaining a one predecessor for 181. This changes the usable helper: in the resulting 210-state construction it is 215 rather than the Research proposal's 129.
3. Move the closed 78-primitive continuation down the right-hand branch chain using twelve more rotations. This makes every source entry to 218 leave a zero immediately to the left, while preserving the other required distinctions. The helper for the 181/220 fold becomes 217.

The final source layout has the same 197 distinct subtrees as all three inputs. The reduction therefore comes from reusing physical decoder states in the available tape context, not from finding a smaller abstract syntax tree.

3.1 A complete rotation recipe

For an outer pure decoder p, inner decoder q, and side s, write (p, q, s). Side zero transforms $((A, B), C)$ to $(A, (B, C))$; side one performs the inverse transformation. A pure decoder preserves its read bit and moves right on both symbols. The inner decoder has exactly the displayed incoming edge. The recorded factors are closed: none of their branch instructions escapes its factor, and they contain no halt. Every new rotation also has only right-moving incoming edges to its two edited decoders; no additional arithmetic entry repair is needed. The existing Contextual repair is retained.

Phase	Sequential triples (p, q, s)		
I	$(215, 204, 1)$	$(217, 181, 1)$	$(185, 217, 0)$
	$(174, 185, 1)$	$(129, 158, 0)$	$(217, 215, 1)$
	$(204, 214, 0)$	$(215, 129, 0)$	
II	$(217, 215, 0)$	$(215, 129, 0)$	$(129, 204, 0)$
	$(204, 214, 0)$		
III	$(162, 203, 0)$	$(203, 223, 0)$	$(223, 188, 0)$
	$(188, 177, 0)$	$(177, 89, 0)$	$(89, 174, 0)$
	$(174, 217, 0)$	$(217, 215, 0)$	$(215, 129, 0)$
	$(129, 204, 0)$	$(204, 214, 0)$	$(214, 216, 0)$

`construction/ledger.json` records every affected source path, all three factor sizes and closure bounds, each row edit, and the metrics after every stage. The independent source auditor reconstructs the paths by literal execution rather than trusting this ledger.

Some intermediate layouts have a run of seven zeros. They are algebraic construction intermediates, not claimed correct final interpreters: seven zeros can be mistaken for the interpreter's origin marker. The delivered layout restores maximum zero run six. The full runtime theorem is checked on the final literal table; no unverified intermediate runtime theorem is used.

3.2 Source control-flow preservation

The source has increment instructions CI_i, conditional decrements CD_i, outward branches CB_k, backward continuations CC_k, and halt CH. Binary addresses describe leaf locations in a full binary syntax tree; the left-to-right leaf order gives source ordinals. CI and positive CD use ordinary successor. Zero CD and CB unwind a specified number of left ancestors; CC unwinds right ancestors. The two implementations of this graph are: subtree intervals with ancestor walking, and arithmetic operations on binary addresses followed by leaf decoding.

Lemma 3.1 (Closed reassociation)**.** *If A, B, C are closed source factors without halt, reassociating $((A, B), C)$ to $(A, (B, C))$ preserves the ordered primitive word and every ordinal control edge, provided the surrounding control bindings remain those checked by the complete source graph.*

Proof. Internal effects and targets of each factor are unchanged because no control instruction escapes the factor. A normal completion proceeds to the next factor in the same order A, B, C. The entire affected subtree occupies the same ordinal interval, so surrounding controls that leave or enter this interval retain their targets. For this particular finite source, both complete graph resolvers verify every edge after every rotation, including external controls and the no-stray-one condition beyond decoded successor leaves. This explicit check closes any ambiguity about the surrounding bindings. □

The construction preserves all 3,909 opcode/operand pairs and all 5,083 edges at each stage. The independent prefix audit checks all 3,908 internal prefixes before and after each of the

24 rotations. The final source therefore runs the same numerical search on the same stores, with a different address representation. This fact alone does not prove correctness of its literal interpreter; Section 5 supplies that separate connection.

4 Local deletion proofs with arbitrary tape exteriors

It is useful to state the joint local proof against a 212-state intermediate P: apply the six Contextual rotations and its engine repair to common U212, then the 24 rotations above, and delay all three deletions. The alternative builder reproduces this P and checks that its final U209 is byte-identical to the primary route. Neither source reassociation nor a finite blank run is used inside the following local proof.

Write the current head coordinate as 0. A state entered only by right-moving transitions that write t has tape bit t at -1 at that entry. This holds for every initial binary tape. None of the tagged states below is the start state. The complete incoming-edge census of P is:

State	Forced left bit	All incoming rows
163	0	214/0
180	1	132/1, 199/1
220	0	204/0
181	1	185/1
218	0	158/0, 211/0, 216/0, 220/0

Every row in the final column writes the forced bit and moves right into the state in the first column. These are global inventories of all physical incoming transitions, not observations from a finite run.

4.1 First fold: $180 \mapsto 163$

In P the relevant states are

q	0	1
163	0 R 90	1 R 211
180	0 R 152	1 R 211
211	0 R 218	1 R 184.

Also $218/0 = 0\,\mathrm{R}\,90$ and $184/0 = 0\,\mathrm{R}\,152$. Redirect every target 180 to 163, remove both rows of 180, and set $163/0 = 0\,\mathrm{L}\,211$.

On current bit one, both old states already have the same one-step action. On current bit zero, the new path first consults the left tag:

$$\begin{aligned} \text{old state } 163, t = 0 &: 163 \to 211 \to 218 \to 90, \\ \text{old state } 180, t = 1 &: 163 \to 211 \to 184 \to 152. \end{aligned}$$

The head displacements are $-1, +1, +1$. Only cells $-1, 0$ are read; both are unchanged, and the head ends at 1, as in the old step. This gives a 211-state intermediate P_1.

4.2 Second fold: $181 \mapsto 220$

In P_1, state 220 has branches $(218, 176)$ and state 181 has branches $(147, 176)$, all preserving the bit and moving right. Their left tags are zero and one respectively. The new common-layout helper is

$$217/0 = 0\,\mathrm{R}\,158, \qquad 217/1 = 1\,\mathrm{R}\,215,$$

with $158/0 = 0\,\mathrm{R}\,218$ and $215/0 = 0\,\mathrm{R}\,147$. Redirect 181 to 220, remove its rows, and set $220/0 = 0\,\mathrm{L}\,217$. The zero-reading paths are

$$220 \to 217 \to 158 \to 218 \quad (t = 0), \qquad 220 \to 217 \to 215 \to 147 \quad (t = 1).$$

Again the entire tape is unchanged, head displacement is $+1$, and the read-one branch takes one step to 176. This gives P_2 with 210 states. In P_2, the complete incoming census for 218 is now just $158/0, 211/0, 216/0$; all still write zero and move right.

4.3 Third fold: $218 \mapsto 157$

Existing state 157 reproduces 218 whenever its left neighbour is zero:

read	old action at 218	replacement path	steps
0	$0\,\mathrm{R}\,90$	$157 \to 67 \to 151 \to 90$	3
1	$1\,\mathrm{R}\,86$	$157 \to 148 \to 186 \to 86$	3.

Both paths preserve every inspected cell at each step. The local checker compares both tape cells as well as the endpoint state and head. Both cases leave the whole tape unchanged and the head at 1. Redirect every target 218 to 157 and remove 218's two rows. No outgoing row of 157 is changed. The result is $M = U209$.

Theorem 4.1 (Joint local halting equivalence)**.** *For every binary tape $T : \mathbb{Z} \to \{0, 1\}$ and every initial head coordinate, P and M, started in state zero on that same tape, halt if and only if each other does.*

Proof. Use the state map $\rho(180) = 163$, $\rho(181) = 220$, $\rho(218) = 157$, with identity elsewhere. At each distinguished source configuration, the preceding incoming row supplies the required left bit. The exact local paths above restore every cell, so an old step has a finite new block ending at the same head and tape and at the mapped state. Every other row is checked to have the identical write, direction, and mapped target. The initial states agree, and halt is fixed by ρ.

The direct composed block for old $163/0$ is

$$163 \to 211 \to 157 \to 67 \to 151 \to 90,$$

which takes five steps. Every other changed block takes three steps; the remaining blocks take one. The checker verifies all ten special state/read cases using only the specified cells $-1, 0$; unspecified exterior cells cannot be read. No intermediate block state is halt. Induction on an old finite run gives a matching new run. If the old run is infinite, positive block lengths yield an infinite, cofinal matching new run. Determinism then excludes an earlier halt in the new computation. This proves both directions. Translation invariance removes the choice of coordinate zero. □

For a halting P run, let $N(q, b)$ count visits to state q reading b, and let $N(218, *)$ count both symbols. The direct proof gives the exact timing identity

$$T_M = T_P + 4N(163, 0) + 2\big(N(180, 0) + N(220, 0) + N(181, 0) + N(218, *)\big),$$

so $T_P \le T_M \le 5T_P$. This is a local-intermediate theorem. It is not a time bound relative to the original U212, whose source addresses have also changed. The construction claims a state saving, not a speedup.

5 Canonical runtime and the blank initializer

The runtime audit is separate from the local proof. It starts each source primitive in a complete canonical frame: a left marker, its binary source address, a zero gap, a header marker, nine primary register blocks, a nonempty finite list of spare register blocks, and infinite zero exterior. A register of value v is encoded as $B(v) = 0\,1^{v+1}$. Logical register i is stored at physical position $\pi(i)$, where $\pi = (6, 2, 0, 4, 3, 7, 8, 5, 1)$. Header distance is $56 + g$ for arbitrary $g \in \mathbb{N}$.

5.1 What the symbolic contracts quantify over

There is one contract for every non-decrement occurrence and two for each conditional decrement, separating zero and positive input. Thus 3,909 occurrences give 5,084 guarded contracts. Each contract uses arbitrary independent natural register values, an arbitrary natural gap, an arbitrary first spare value, and an arbitrary finite remaining spare list. They are not 5,084 small numerical tests.

The checker represents unary lengths by nonnegative affine expressions and the remaining list by an opaque list atom. Its mathematical rules are:

1. A literal transition reads a known bit, writes the table's bit, and updates both the state and the oriented tape decomposition.
2. A same-state unary scan is justified by induction on its arbitrary natural length, including length zero.
3. Four list-traversal rules are rederived from the exact table on one arbitrary block with opaque surrounding contexts, then lifted to finite lists by base/cons induction.
4. Word normalization uses concatenation of adjacent runs and the verified identities $C(A)1 = 1B(A)$ and their reversals, where $C(v) = 1\,0\,1^{v}$. It preserves the entire represented tape.

Every case ends either in the canonical successor frame with the required register effect, or at literal halt for CH. Its duration is a nonnegative affine expression with a strictly positive constant. This is the progress condition needed when concatenating the primitive simulations into infinite computations. All 418 literal rows occur in the fresh derivations, including the one-block proofs.

The delivered replay completed all 5,084 contracts using 1,250,049 symbolic inference applications. The minimum duration constant was 5. The full case records, checker hashes, and exact table/source hashes are included. A standalone C++ interpreter independently passed 15,252 finite primitive cases, taking 6,773,382 literal steps in that fixed suite. These finite checks exercise a different implementation; they do not replace the quantified argument.

The symbolic inference engine is ordinary Python, not a proof-assistant kernel. Assertions must remain enabled for its replay, and the wrapper rejects optimized Python. The rules above and their induction proofs are part of the mathematical trust boundary. Three successful replays using inherited variants of the same engine do not remove that boundary.

5.2 Exact initialization and source seed

The 209-state table has the same complete 48,140-transition blank trace as common U212 and Contextual U211. No changed row is touched in this prefix. Its endpoint is native state 84, head -2, header coordinate 54, and source ordinal 136 with address `001100110101010`. In logical order $(N, T, W, A, D, H, X, Y, L)$, the store is

$$(1, 0, 13, 0, 0, 1, 0, 0, 1).$$

There are 43 spare registers of value one, and exactly 120 one-cells in the complete tape. All other cells are zero. The check compares the whole finite support and the full transition-trace hash.

This is not a clean-root initial source configuration. The retained source bridge executes 141 primitives from ordinal 106 and store $(1, 0, 4, 0, 0, 1, 0, 0, 1)$ to obtain this endpoint. Because ordinal source control flow is unchanged, that bridge transports to the new source. The inherited remaining axiom-builder suffix produces nine explicit theorem heads above opaque Cantor tail 4. The cache value 1 in the prefix is an arithmetic selector parameter, not an assumed proof of falsity. The scheduler's worst prefix debt is two, leaving a permanent seven-head margin that keeps the tail inaccessible. An explicit ghost list length handles the alias $P(0, 0) = 0$.

The retained strong compiler operates over arbitrary underlying stacks, successful candidate indices are cofinal, and unsuccessful candidate computations terminate. These completeness facts are needed in addition to soundness: a machine that never finds a false proof for an incomplete search would not give the desired consistency biconditional. The original mathematical arguments and their full archive dependencies are preserved.

6 The theory U and the consistency endpoint

The unary-theory archive uses classical first-order logic on nonempty domains, with one binary relation E, no primitive equality, functions, or constants. Put

$$\mathrm{Sub}(x, a) \ :\Longleftrightarrow\ \forall t\,(E(t, x) \to E(t, a)).$$

For every formula $\varphi(x)$ whose free variables are contained in $\{x\}$, the only nonlogical schema of U is

$$\forall a\,\exists b\,\left[\forall x\,(E(x, b) \to \varphi(x))\ \wedge\ \forall x\,(\mathrm{Sub}(x, a) \to (\varphi(x) \to E(x, b)))\right].$$

Bound variables are freshened; in particular a is not a permitted free matrix parameter. Arbitrarily many bound variables remain available, so this is not one-variable first-order logic. The two clauses specify a weak cover; they do not impose ambient extensionality or an exact power filter [4].

6.1 Review of the inherited PA calibration

The supplied lower interpretation uses numerical representatives modulo coextensionality. Coextensionality is automatically a congruence in the containing-object coordinate of E, whereas the member coordinate needs a saturated container. The numerical predicate includes this saturation, transitivity, strict membership linearity on classes, saturation of members, and least/greatest endpoint conditions for intersections with existing objects.

The order of the construction matters. Local predecessor facts first prove full induction without assuming that every numerical object already has a successor. Formula parameters are bound inside a legal unary matrix, for example

$$Q(v) = \forall \bar{p}\,(\mathrm{BaseAndStep}(\theta, \bar{p}) \to \theta(v, \bar{p})).$$

Comparison and least numerical failures then yield bounded numerical comprehension. The proof chooses a lexicographically least failing numerical tuple and binds that tuple inside the unary matrix. A support source can be chosen externally in the proof without becoming a free parameter of the schema instance. Only after this step are total successors, numerical powers, pair codes, graph histories, addition, and multiplication derived. History uniqueness is established before history extension, so no illicit free graph parameter enters the filter formula.

In the upper direction, PA interprets $E(x, a)$ as the x-th bit of a. PA proves that inclusion of the one-bits of x among those of a implies $x \leq a$. For each fixed arithmetic formula, ordinary induction constructs its finite bitset below $a + 1$, giving the required cover. This does not define a uniform truth predicate. It also interprets the stronger cover schema with parameters, denoted W, and global irreflexivity IR.

The supplied proof claims uniform primitive-recursive translations of finite refutations, whose correctness is arithmetically proved by induction on syntax and proof sequences. This is what supports the internal statement

$$\mathsf{PA} \vdash \mathrm{Con}(\mathsf{U}) \leftrightarrow \mathrm{Con}(\mathsf{PA}),$$

rather than only external semantic equiconsistency. The independent audit found no concrete error in this chain. It did not implement an arbitrary-proof translator or export a proof-assistant derivation; those remain limitations of the inherited calibration.

6.2 What the literal source actually searches

The source is the inherited numerical calculus $\mathcal{D}_U$, with Cantor coding, truth alias zero, falsity one, seven logical rows, a dominated propositional row, modus ponens, generalization, and the gated cover/vacuity bundle. Its cover permits additional matrix parameters and is calibrated through $\mathsf{W} + \mathsf{IR}$. It is not literally the parameter-free schema above or the separate positive dyadic emitter also proposed in the unary-theory manuscript.

The retained encoding proofs handle junk codes, variable renaming, freshness, and colliding labels. In particular, a matrix code also serves as an output variable label because every effectively used label of a positive code is smaller than that code. Code zero is a closed truth alias. The gate excluding the tested label from the source and output labels is substantive: deleting it would make the bundled atomic-vacuity row invalid. The exceptional source/output coincidence is handled by an empty source under irreflexivity. Completeness pads a matrix to a suitable positive code and restores variable names using finite logical templates.

Source preservation keeps all these arguments applicable. The intended composition is

$$\mathsf{PA} \vdash \neg\,\mathrm{Halt}(M, \mathrm{blank}) \;\leftrightarrow\; \mathrm{Con}(\mathcal{D}_U) \;\leftrightarrow\; \mathrm{Con}(\mathsf{W} + \mathsf{IR}) \;\leftrightarrow\; \mathrm{Con}(\mathsf{U}) \;\leftrightarrow\; \mathrm{Con}(\mathsf{PA}).$$

For readability this display expresses the chain of biconditionals; it is not asserting that the literal axiom sets are equal. Each bridge has its own finite-proof or machine-simulation argument. The formal K215 theorem about ZFC is important background, but changing its machine and theory does not transfer its MM0 certificate to this endpoint [5, 6].

6.3 A small obstruction to a tempting theory shortening

One might remove U's global upper clause and keep only local exactness:

$$\forall a\, \exists b\, \forall x\, [\mathrm{Sub}(x, a) \rightarrow (E(x, b) \leftrightarrow \varphi(x))]. \tag{L}$$

This change admits a two-object model. Let the domain be $\{0, 1\}$ and let E be the off-diagonal relation. The two membership profiles are distinct singletons, so $\mathrm{Sub}(x, a)$ holds precisely when $x = a$. For any a, choose $b = a$ if $\varphi(a)$ is false and $b \neq a$ if it is true. Then (L) holds, even if additional parameters are allowed in φ. The finite-model construction and its soundness proof are formalizable in PA, which therefore proves consistency of this weakened schema. Assuming PA is consistent, PA cannot also prove the required equivalence with its own consistency. Thus the apparently simpler local filter loses the intended calibration. The global upper clause performs essential work. This observation helps explain why the present successful reduction changes the physical decoder while preserving the logical source.

7 Evidence, reproduction, and remaining work

Obligation	Fresh result
Construction	Both U211/24-rotation and U212/30-rotation routes give the same exact 209-state table.
Source	Every primitive, operand, ordinal edge, and all internal prefix bindings checked; final maximum zero run six.
Local folds	Exact incoming inventories, ten universal direct cases, all remaining row identities, and nine rejected corruptions; opaque exterior never read.
Runtime	5,084 arbitrary-natural cases; 418/418 row coverage; 1,250,049 inference applications.
Blank startup	Identical complete 48,140-step trace, 120 one-cells, and 141-step source-prefix bridge.
Independent finite execution	15,252 primitive cases; 6,773,382 literal steps in the fixed suite.
Input comparison	All three supplied U211 full replays passed; exact source equality and rooted non-isomorphism checked.
Formal endpoint	Mathematical PA-formalizability argument with inherited calibration; no new Lean, Rocq, MM0, or exported PA proof object.

The archive is self-contained for the included standard-library Python checks. From its root, run:

```
python3 -B reproduce.py --quick
python3 -B reproduce.py
```

The quick mode explicitly skips fresh unbounded symbolic derivation. It checks hashes, reconstructs both routes, compares the frozen outputs, audits rotations and local proofs, replays startup, and runs targeted negative controls. The default additionally rederives all 5,084 symbolic cases and compares their records exactly. It does not merely trust saved summaries. Do not use `-O` or `PYTHONOPTIMIZE`. Regenerated results are written under a temporary directory, leaving the sealed inputs and evidence intact.

For the independent finite executor, a C++17 compiler suffices:

```
g++ -std=c++17 -O3 code/check_finite_runtime.cpp -o /tmp/check_u209
/tmp/check_u209 machine/u209.native.tm machine/sites.tsv
```

The supplied input archives retain their original complete reproduction scripts and proof dependencies. The comparison receipt records fresh runs of those scripts; repeating all historical checks is optional and is not silently included in a quick run.

The report is rebuilt from editable LaTeX with two passes:

```
pdflatex -interaction=nonstopmode -halt-on-error report.tex
pdflatex -interaction=nonstopmode -halt-on-error report.tex
```

PDF bytes can vary with TeX versions and timestamps. Check the delivered manifest before rebuilding, or compile a separate copy. The mathematical identity of the machine is its table, not the PDF metadata.

The next substantive verification target is an exported end-to-end PA or proof-assistant endpoint for the exact 209-state table and the stated numerical theory. Further state minimization would require another compatible tape-context reuse or a redesigned source/runtime interface. No lower bound is proved here. The present result is a concrete two-state improvement over every supplied 211-state candidate, with its new obligations made explicit and reproducible.

A Exact retained-row patch from Contextual U211

All native transitions absent from this table retain their action, except that the outgoing rows of native 181 and 218 are removed. The table includes every changed target, so it is a complete patch. Native halt is 254. The source-address recipe must accompany this literal patch when reproducing the proof.

Native state	Read	Old action	New action
89	0	$0R155$	$0R185$
129	0	$0R158$	$0R146$
129	1	$1R178$	$1R204$
158	0	$0R98$	$0R157$
158	1	$1R218$	$1R178$
162	0	$0R203$	$0R207$
162	1	$1R116$	$1R203$
174	0	$0R159$	$0R98$
174	1	$1R185$	$1R217$
177	0	$0R132$	$0R155$
185	0	$0R217$	$0R159$
185	1	$1R215$	$1R220$
188	0	$0R153$	$0R132$
203	0	$0R207$	$0R67$
204	1	$1R216$	$1R214$
211	0	$0R218$	$0R157$
214	0	$0R147$	$0R163$
214	1	$1R146$	$1R216$
215	0	$0R214$	$0R147$
215	1	$1R204$	$1R129$
216	0	$0R163$	$0R157$
216	1	$1R218$	$1R116$
217	0	$0R147$	$0R158$
217	1	$1R181$	$1R215$
220	0	$0R218$	$0L217$
223	0	$0R67$	$0R153$

B Complete dense transition table

There are exactly 209 working-state entries below, each giving its two transitions. An entry wDr means write w, move in direction D, and enter r. Label 209 is halt and has no row.

State	Read 0	Read 1	State	Read 0	Read 1
0	$1R92$	$1R0$	105	$0R21$	$0L10$
1	$0L1$	$1L2$	106	$0R61$	$1R42$
2	$0L3$	$1L2$	107	$0R121$	$1R8$
3	$0L5$	$1L2$	108	$0R101$	$1R122$
4	$1R6$	$0L4$	109	$0L208$	$0L95$
5	$0L9$	$1L2$	110	$0R124$	$1R104$
6	$0R6$	$0R10$	111	$0R115$	$1L72$
7	$0R190$	$1R159$	112	$0R165$	$1R181$
8	$0R14$	$1R13$	113	$0R160$	$1R184$
9	$0L15$	$1L2$	114	$0R148$	$1R146$
10	$0L16$	$0R10$	115	$0R11$	$1R129$
11	$0L26$	$1R128$	116	$0L131$	$0L1$
12	$0R19$	$1R20$	117	$0R39$	$1R72$
13	$0L89$	$1L31$	118	$0R61$	$1R132$
14	$0L11$	$0R22$	119	$0R61$	$1R133$
15	$0L23$	$1L2$	120	$0R70$	$0L16$
16	$1L1$	$0L1$	121	$0R14$	$1R132$
17	$0R24$	$0R32$	122	$0R21$	$1R134$
18	$0R98$	$1R113$	123	$0R139$	$1R189$
19	$0R28$	$1R29$	124	$0R137$	$1L20$
20	$0R30$	$1R31$	125	$0R130$	$1R180$
21	$0R24$	$0R59$	126	$0R204$	$1R156$
22	$0R33$	$1R22$	127	$0R75$	$1R176$
23	$0L34$	$1L2$	128	$0R75$	$1R206$
24	$0L35$	$0R0$	129	$0R203$	$1R157$
25	$0R36$	$1R25$	130	$0R12$	$1R141$
26	$0R37$	$1R38$	131	$0R30$	$1R69$
27	$0R39$	$0L40$	132	$0R39$	$1L124$
28	$0R41$	$1R42$	133	$0R153$	$0L16$
29	$0R43$	$1R44$	134	$0R154$	$0L16$
30	$0R45$	$1R46$	135	$0R88$	$1R136$
31	$0R21$	$1R47$	136	$0R197$	$1R140$
32	$0R48$	$1R32$	137	$0R24$	$0R48$
33	$0R32$	$1R33$	138	$0R99$	$1R37$
34	$0L74$	$1L2$	139	$0R172$	$1R148$
35	$1R35$	$1L1$	140	$0R96$	$1R159$
36	$0R50$	$1R36$	141	$0R175$	$1R166$
37	$0R17$	$1R111$	142	$0R12$	$1R159$
38	$0R13$	$1R52$	143	$0R88$	$1L166$
39	$0L11$	$0R36$	144	$0R88$	$1R158$
40	$0L4$	$1L53$	145	$0L152$	$1L163$
41	$0R24$	$0R25$	146	$0R84$	$1R88$
42	$0R13$	$0L16$	147	$0R159$	$1R12$
43	$0R55$	$1R56$	148	$0R83$	$1R143$
44	$0R57$	$1R58$	149	$0R144$	$1R155$
45	$0R24$	$0R54$	150	$0L66$	$1L141$
46	$0R60$	$1R61$	151	$0R150$	$1R170$
47	$0R62$	$0L16$	152	$0R159$	$1R114$
48	$0R25$	$1R48$	153	$0R97$	$1R97$
49	$0L35$	$0R63$	154	$0R49$	$0R32$
50	$0R16$	$1R50$	155	$0R192$	$1R188$

State	Read 0	Read 1	State	Read 0	Read 1
51	$0R147$	$1R143$	156	$0L195$	$1R195$
52	$0R67$	$1R68$	157	$0L187$	$1R96$
53	$0L40$	$1L53$	158	$0R168$	$1R179$
54	$0R22$	$1R54$	159	$0R19$	$1R171$
55	$0L11$	$0R50$	160	$0R164$	$1R187$
56	$0R60$	$0L16$	161	$0R159$	$1R96$
57	$0R69$	$1R70$	162	$0R125$	$1R164$
58	$0R61$	$1R71$	163	$0R97$	$1R162$
59	$0R54$	$1R59$	164	$0L190$	$1R142$
60	$0R49$	$0R25$	165	$0L66$	$1R173$
61	$0L28$	$1L72$	166	$0R146$	$1R172$
62	$0L117$	$1L30$	167	$0R96$	$1R201$
63	$0R73$	$1R63$	168	$0R161$	$1R143$
64	$0R97$	$1R150$	169	$0R148$	$1R87$
65	$1R185$	$1L53$	170	$0R161$	$1R96$
66	$0R144$	$1R145$	171	$0R181$	$1R122$
67	$0R76$	$1R209$	172	$0L193$	$1R60$
68	$0L197$	$1R31$	173	$0R145$	$1R88$
69	$0R41$	$1R62$	174	$0R152$	$1R202$
70	$0L89$	$1R78$	175	$0R140$	$1R84$
71	$0R55$	$1L79$	176	$0R175$	$1R75$
72	$0R80$	$1R81$	177	$0R126$	$1R169$
73	$1R0$	$1R63$	178	$0R138$	$1R116$
74	$0R74$	$1R82$	179	$0R97$	$1R51$
75	$0L145$	$1L141$	180	$0R168$	$1R51$
76	$0R85$	$1R86$	181	$0R17$	$1L131$
77	$0R20$	$1R38$	182	$0R143$	$1R149$
78	$0R89$	$1L132$	183	$0R76$	$1R116$
79	$0L108$	$1R56$	184	$0R106$	$1R124$
80	$0R49$	$0R33$	185	$1L40$	$1L109$
81	$0R91$	$0L42$	186	$0R175$	$1R156$
82	$0R178$	$1R77$	187	$0R182$	$1R183$
83	$0R96$	$1R12$	188	$0R66$	$1R205$
84	$0R88$	$1R97$	189	$0R202$	$1R198$
85	$0R207$	$1R21$	190	$0R84$	$1R191$
86	$0R116$	$1R100$	191	$0R157$	$1R186$
87	$0R174$	$1R167$	192	$0R64$	$1R7$
88	$0R102$	$1R103$	193	$0R150$	$1R127$
89	$0R80$	$1R79$	194	$0R18$	$1R94$
90	$0R101$	$1R196$	195	$0R150$	$1R173$
91	$0R105$	$1R106$	196	$0R104$	$1R96$
92	$0R65$	$0R0$	197	$0R90$	$1R88$
93	$0R27$	$1R107$	198	$0R156$	$1R200$
94	$0R183$	$0L208$	199	$0R140$	$1R123$
95	$1L109$	$1L95$	200	$0R150$	$1R112$
96	$0R19$	$1R110$	201	$0R151$	$1R199$
97	$0R60$	$1R60$	202	$0L201$	$1R168$
98	$0R68$	$1R135$	203	$0R197$	$1R28$
99	$0R93$	$1R108$	204	$0R163$	$1R193$
100	$0R105$	$0L208$	205	$0R146$	$1R177$
101	$0R117$	$1R118$	206	$0R161$	$1R194$
102	$0R71$	$1R58$	207	$0R45$	$1R100$
103	$0R57$	$1R119$	208	$0L4$	$0L208$
104	$0R21$	$1R120$			

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[1] Supplied research packet, *U211: a 211-state candidate for the consistency of Peano arithmetic*, 28 October 2026. File `U211_ConPA_Contextual_2026-10-28.zip`, ProveIt, pinned commit stated in Section 2; exact archive retained in `inputs/`.

[2] Supplied research packet, *U211: a decoder-sharing candidate for Con(PA)*, 28 October 2026. File `U211_ConPA_Research_2026-10-28.zip`; internal directory `U211_ConPA_Decoder_Reassociation_2026-10-28`. Exact archive retained in `inputs/`.

[3] Supplied research packet, *U211: decoder reassociation and a guarded state deletion*, 28 October 2026. File `U211_ConPA_Rotated_2026-10-28.zip`. Exact archive retained in `inputs/`.

[4] Supplied research packet, *Unary power covering at PA strength: Eliminating the source parameter, with a division-free positive proof-search interface*, 27 October 2026, `pa_unary_power_cover.zip`. The complete paper and implementation artifacts are retained in `inputs/`; see especially its unary-core, unary-arithmetic, upper-transport, and positive-interface sections.

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[7] Jeff's busy-beaver reference development, `ZF/m254`, branch `zf-254`. https://github.com/jeff87654/busy-beaver/tree/zf-254/ZF/m254. Upstream provenance is preserved by the retained input packets and license notices.